

Inversion

■ Nonsingular Matrices

We now come to a special type of square matrix and formulate the notion corresponding to the reciprocal of a nonzero real number.

An $n \times n$ matrix A is called **nonsingular**, or **invertible**, if there exists an $n \times n$ matrix B such that $AB = BA = I_n$; such a B is called an **inverse** of A . Otherwise, A is called **singular**, or **noninvertible**.

Let $A = \begin{bmatrix} 2 & 3 \\ 2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & \frac{3}{2} \\ 1 & -1 \end{bmatrix}$. Since $AB = BA = I_2$, we conclude that B is an inverse of A . ■

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Proof

Let B and C be inverses of A . Then

$$AB = BA = I_n \quad \text{and} \quad AC = CA = I_n.$$

We then have $B = BI_n = B(AC) = (BA)C = I_n C = C$, which proves that the inverse of a matrix, if it exists, is unique. 

Because of this uniqueness, we write the inverse of a nonsingular matrix A as A^{-1} . Thus

$$AA^{-1} = A^{-1}A = I_n.$$

EXAMPLE

Let

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

If A^{-1} exists, let

$$A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Then we must have

$$AA^{-1} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

so that

$$\begin{bmatrix} a + 2c & b + 2d \\ 3a + 4c & 3b + 4d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Equating corresponding entries of these two matrices, we obtain the linear systems

$$\begin{array}{lcl} a + 2c = 1 & \text{and} & b + 2d = 0 \\ 3a + 4c = 0 & & 3b + 4d = 1. \end{array}$$

The solutions are (verify) $a = -2$, $c = \frac{3}{2}$, $b = 1$, and $d = -\frac{1}{2}$. Moreover, since the matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

also satisfies the property that

$$\begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

we conclude that A is nonsingular and that

$$A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}.$$

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$$\begin{array}{lcl} a + 2c = 1 & \text{and} & b + 2d = 0 \\ 2a + 4c = 0 & & 2b + 4d = 1. \end{array}$$

These linear systems have no solutions, so our assumption that A^{-1} exists is incorrect. Thus A is singular. ■

Finding A^{-1}

for A nonsingular, we transform the partitioned matrix $[A \mid I_n]$ to reduced row echelon form, obtaining $[I_n \mid A^{-1}]$.

Let

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 5 & 5 & 1 \end{bmatrix}.$$

Assuming that A is nonsingular, we form

$$[A \mid I_3] = \left[\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 2 & 3 & 0 & 1 & 0 \\ 5 & 5 & 1 & 0 & 0 & 1 \end{array} \right].$$

We now perform elementary row operations that transform $[A \mid I_3]$ to $[I_3 \mid A^{-1}]$; we consider $[A \mid I_3]$ as a 3×6 matrix, and whatever we do to a row of A we also do to the corresponding row of I_3 . In place of using elementary matrices directly, we arrange our computations, using elementary row operations as follows:

	A				I_3			
1	1	1		1	0	0	Apply $-5\mathbf{r}_1 + \mathbf{r}_3 \rightarrow \mathbf{r}_3$.	
0	2	3		0	1	0		
5	5	1		0	0	1		

1	1	1		1	0	0	Apply $\frac{1}{2}\mathbf{r}_2 \rightarrow \mathbf{r}_2$.
0	2	3		0	1	0	
0	0	-4		-5	0	1	

1	1	1		1	0	0	Apply $-\frac{1}{4}\mathbf{r}_3 \rightarrow \mathbf{r}_3$.
0	1	$\frac{3}{2}$		0	$\frac{1}{2}$	0	
0	0	-4		-5	0	1	

$$\begin{array}{ccc|ccc}
 1 & 1 & 1 & 1 & 0 & 0 \\
 0 & 1 & \frac{3}{2} & 0 & \frac{1}{2} & 0 \\
 0 & 0 & 1 & \frac{5}{4} & 0 & -\frac{1}{4}
 \end{array}$$

Apply $-\frac{3}{2}\mathbf{r}_3 + \mathbf{r}_2 \rightarrow \mathbf{r}_2$ and
 $-\mathbf{r}_3 + \mathbf{r}_1 \rightarrow \mathbf{r}_1$.

$$\begin{array}{ccc|ccc}
 1 & 1 & 0 & -\frac{1}{4} & 0 & \frac{1}{4} \\
 0 & 1 & 0 & -\frac{15}{8} & \frac{1}{2} & \frac{3}{8} \\
 0 & 0 & 1 & \frac{5}{4} & 0 & -\frac{1}{4}
 \end{array}$$

Apply $-\mathbf{r}_2 + \mathbf{r}_1 \rightarrow \mathbf{r}_1$.

$$\begin{array}{ccc|ccc}
 1 & 0 & 0 & \frac{13}{8} & -\frac{1}{2} & -\frac{1}{8} \\
 0 & 1 & 0 & -\frac{15}{8} & \frac{1}{2} & \frac{3}{8} \\
 0 & 0 & 1 & \frac{5}{4} & 0 & -\frac{1}{4}
 \end{array}$$

Hence

$$A^{-1} = \begin{bmatrix} \frac{13}{8} & -\frac{1}{2} & -\frac{1}{8} \\ -\frac{15}{8} & \frac{1}{2} & \frac{3}{8} \\ \frac{5}{4} & 0 & -\frac{1}{4} \end{bmatrix}.$$

We can readily verify that $AA^{-1} = A^{-1}A = I_3$. ■

The question that arises at this point is how to tell when A is singular. The answer is that A is singular if and only if A is row equivalent to matrix B , having at least one row that consists entirely of zeros. We now prove this result.

Theorem

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This means that in order to find A^{-1} , we do not have to determine, in advance, whether it exists. We merely start to calculate A^{-1} ; if at any point in the computation we find a matrix B that is row equivalent to A and has a row of zeros, then A^{-1} does not exist. That is, we transform the partitioned matrix $[A \mid I_n]$ to reduced row echelon form, obtaining $[C \mid D]$. If $C = I_n$, then $D = A^{-1}$. If $C \neq I_n$, then C has a row of zeros and we conclude that A is singular.

Let

$$A = \begin{bmatrix} 1 & 2 & -3 \\ 1 & -2 & 1 \\ 5 & -2 & -3 \end{bmatrix}.$$

find A^{-1}

Theorem 1: A square matrix has an inverse if and only if its determinant is not zero.

Below we develop a method to calculate the inverse of nonsingular matrices using determinants.

Definition 1: The **cofactor matrix** associated with an $n \times n$ matrix $\mathbf{A}$ is an $n \times n$ matrix $\mathbf{A}^c$ obtained from $\mathbf{A}$ by replacing each element of $\mathbf{A}$ by its cofactor.

Definition 2: The **adjugate(adjoint)** of an $n \times n$ matrix $\mathbf{A}$ is the transpose of the cofactor matrix of $\mathbf{A}$: $\mathbf{A}^* = (\mathbf{A}^c)^T$

Let $A = [a_{ij}]$ be an $n \times n$ matrix. The $n \times n$ matrix $\text{adj } A$, called the **adjoint** of A , is the matrix whose (i, j) th entry is the cofactor A_{ji} of a_{ji} . Thus

$$\text{adj } A = \begin{bmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \vdots & \vdots & & \vdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{bmatrix}.$$

Remark It should be noted that the term *adjoint* has other meanings in linear algebra in addition to its use in the preceding definition.

Let $A = \begin{bmatrix} 3 & -2 & 1 \\ 5 & 6 & 2 \\ 1 & 0 & -3 \end{bmatrix}$. Compute $\text{adj } A$.

Solution

We first compute the cofactors of A . We have

$$A_{11} = (-1)^{1+1} \begin{vmatrix} 6 & 2 \\ 0 & -3 \end{vmatrix} = -18,$$

$$A_{12} = (-1)^{1+2} \begin{vmatrix} 5 & 2 \\ 1 & -3 \end{vmatrix} = 17, \quad A_{13} = (-1)^{1+3} \begin{vmatrix} 5 & 6 \\ 1 & 0 \end{vmatrix} = -6,$$

$$A_{21} = (-1)^{2+1} \begin{vmatrix} -2 & 1 \\ 0 & -3 \end{vmatrix} = -6,$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 3 & 1 \\ 1 & -3 \end{vmatrix} = -10, \quad A_{23} = (-1)^{2+3} \begin{vmatrix} 3 & -2 \\ 1 & 0 \end{vmatrix} = -2,$$

$$A_{31} = (-1)^{3+1} \begin{vmatrix} -2 & 1 \\ 6 & 2 \end{vmatrix} = -10,$$

$$A_{32} = (-1)^{3+2} \begin{vmatrix} 3 & 1 \\ 5 & 2 \end{vmatrix} = -1, \quad A_{33} = (-1)^{3+3} \begin{vmatrix} 3 & -2 \\ 5 & 6 \end{vmatrix} = 28.$$

Then

$$\operatorname{adj} A = \begin{bmatrix} -18 & -6 & -10 \\ 17 & -10 & -1 \\ -6 & -2 & 28 \end{bmatrix}.$$

Example of finding adjoint

- Find the adjoint of

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}$$

Solution:

The cofactor matrix of A:

Example of finding adjoint

- Find the adjoint of

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}$$

Solution:

The cofactor matrix of A:

$$\begin{bmatrix} \begin{vmatrix} -2 & 1 \\ 0 & -2 \end{vmatrix} & -\begin{vmatrix} 0 & 1 \\ 1 & -2 \end{vmatrix} & \begin{vmatrix} 0 & -2 \\ 1 & 0 \end{vmatrix} \\ -\begin{vmatrix} 3 & 2 \\ 0 & -2 \end{vmatrix} & \begin{vmatrix} -1 & 2 \\ 1 & -2 \end{vmatrix} & -\begin{vmatrix} -1 & 3 \\ 1 & 0 \end{vmatrix} \\ \begin{vmatrix} 3 & 2 \\ -2 & 1 \end{vmatrix} & -\begin{vmatrix} -1 & 2 \\ 0 & 1 \end{vmatrix} & \begin{vmatrix} -1 & 3 \\ 0 & -2 \end{vmatrix} \end{bmatrix} = \begin{bmatrix} 4 & 1 & 2 \\ 6 & 0 & 3 \\ 7 & 1 & 2 \end{bmatrix}$$
$$A^* = \begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix}$$

Theorem

If $A = [a_{ij}]$ is an $n \times n$ matrix, then $A(\text{adj } A) = (\text{adj } A)A = \det(A)I_n$.

$$\begin{aligned} \begin{bmatrix} 3 & -2 & 1 \\ 5 & 6 & 2 \\ 1 & 0 & -3 \end{bmatrix} \begin{bmatrix} -18 & -6 & -10 \\ 17 & -10 & -1 \\ -6 & -2 & 28 \end{bmatrix} &= \begin{bmatrix} -94 & 0 & 0 \\ 0 & -94 & 0 \\ 0 & 0 & -94 \end{bmatrix} \\ &= -94 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

and

$$\begin{bmatrix} -18 & -6 & -10 \\ 17 & -10 & -1 \\ -6 & -2 & 28 \end{bmatrix} \begin{bmatrix} 3 & -2 & 1 \\ 5 & 6 & 2 \\ 1 & 0 & -3 \end{bmatrix} = -94 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad \blacksquare$$

We now have a new method for finding the inverse of a nonsingular matrix, and we state this result as the following corollary:

Inversion using determinants

$$A^{-1} = \frac{1}{|A|} A^* \quad \text{if } |A| \neq 0$$

That is, if $|A| \neq 0$, then A^{-1} may be obtained by dividing the adjugate of A by the determinant of A .

For example, if

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

then

$$A^{-1} = \frac{1}{|A|} A^* = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Inversion using determinants: example

$$A = \begin{bmatrix} -1 & 3 & 2 \\ 0 & -2 & 1 \\ 1 & 0 & -2 \end{bmatrix}$$

$$|A| = (-1)(-2)(-2) + (3)(1)(1) - (1)(-2)(2) = 3$$

Use the adjugate of

to find A^{-1}

$$A^a = A^* = \begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} A^* = \frac{1}{3} \begin{bmatrix} 4 & 6 & 7 \\ 1 & 0 & 1 \\ 2 & 3 & 2 \end{bmatrix} = \begin{bmatrix} \frac{4}{3} & 2 & \frac{7}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{2}{3} & 1 & \frac{2}{3} \end{bmatrix}$$

More on Invertible Matrices

THEOREM If A and B are invertible matrices with the same size, then AB is invertible and

$$(AB)^{-1} = B^{-1}A^{-1}$$

THEOREM If A is invertible and n is a nonnegative integer, then:

- (a) A^{-1} is invertible and $(A^{-1})^{-1} = A$.
- (b) A^n is invertible and $(A^n)^{-1} = A^{-n} = (A^{-1})^n$.
- (c) kA is invertible for any nonzero scalar k , and $(kA)^{-1} = k^{-1}A^{-1}$.

THEOREM If A is an invertible matrix, then A^T is also invertible and

$$(A^T)^{-1} = (A^{-1})^T$$

Theorem

If A and B are both nonsingular $n \times n$ matrices, then AB is nonsingular and $(AB)^{-1} = B^{-1}A^{-1}$.

Proof

We have $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = (AI_n)A^{-1} = AA^{-1} = I_n$. Similarly, $(B^{-1}A^{-1})(AB) = I_n$. Therefore AB is nonsingular. Since the inverse of a matrix is unique, we conclude that $(AB)^{-1} = B^{-1}A^{-1}$. 

Corollary

If $A_1, A_2, \dots, A_r$ are $n \times n$ nonsingular matrices, then $A_1 A_2 \cdots A_r$ is nonsingular and $(A_1 A_2 \cdots A_r)^{-1} = A_r^{-1} A_{r-1}^{-1} \cdots A_1^{-1}$.

Theorem

If A is a nonsingular matrix, then A^{-1} is nonsingular and $(A^{-1})^{-1} = A$.

Theorem

If A is a nonsingular matrix, then A^T is nonsingular and $(A^{-1})^T = (A^T)^{-1}$.

Theorem

If A is a nonsingular matrix, then A^T is nonsingular and $(A^{-1})^T = (A^T)^{-1}$.

Proof

We have $AA^{-1} = I_n$. Taking transposes of both sides, we get

$$(A^{-1})^T A^T = I_n^T = I_n.$$

Taking transposes of both sides of the equation $A^{-1}A = I_n$, we find, similarly, that

$$(A^T)(A^{-1})^T = I_n.$$

These equations imply that $(A^{-1})^T = (A^T)^{-1}$. 

If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},$$

$$A^{-1} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix} \quad \text{and} \quad (A^{-1})^T = \begin{bmatrix} -2 & \frac{3}{2} \\ 1 & -\frac{1}{2} \end{bmatrix}.$$

Also (verify),

$$A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \quad \text{and} \quad (A^T)^{-1} = \begin{bmatrix} -2 & \frac{3}{2} \\ 1 & -\frac{1}{2} \end{bmatrix}.$$

Solving Linear Systems

Linear Systems and Inverses

If A is an $n \times n$ matrix, then the linear system $A\mathbf{x} = \mathbf{b}$ is a system of n equations in n unknowns. Suppose that A is nonsingular. Then A^{-1} exists, and we can multiply $A\mathbf{x} = \mathbf{b}$ by A^{-1} on the left on both sides, yielding

$$\begin{aligned}A^{-1}(A\mathbf{x}) &= A^{-1}\mathbf{b} \\(A^{-1}A)\mathbf{x} &= A^{-1}\mathbf{b} \\I_n\mathbf{x} &= A^{-1}\mathbf{b} \\ \mathbf{x} &= A^{-1}\mathbf{b}.\end{aligned}\tag{2}$$

Moreover, $\mathbf{x} = A^{-1}\mathbf{b}$ is clearly a solution to the given linear system. Thus, if A is nonsingular, we have a unique solution. We restate this result for emphasis:

If A is an $n \times n$ matrix, then the linear system $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$. Moreover, if $\mathbf{b} = \mathbf{0}$, then the unique solution to the homogeneous system $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$.

1. For each of the following matrices, find the inverse or show that it does not exist. In the latter case, check by calculating the determinant.

$$\text{a) } \begin{bmatrix} 1 & 1 & -1 \\ 2 & -1 & 1 \\ 1 & 1 & 2 \end{bmatrix} \quad \text{b) } \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\text{c) } \begin{bmatrix} 1 & -1 & -1 \\ 2 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix} \quad \text{d) } \begin{bmatrix} 2 & 3 & 1 \\ -1 & 2 & 1 \\ 4 & -1 & -1 \end{bmatrix}$$

2. Use the method of row reduction to evaluate the following determinants:

$$\text{a) } \begin{vmatrix} 1 & 4 & 4 & 1 \\ 0 & 1 & -2 & 2 \\ 3 & 3 & 1 & 4 \\ 0 & 1 & -3 & -2 \end{vmatrix} \quad \text{b) } \begin{vmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & -2 & 0 \\ -2 & 3 & -2 & 3 \\ 0 & -3 & 3 & 3 \end{vmatrix}$$

For #1-6, compute the determinant of the given matrix.

1.)

$$\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$$

2.)

$$\begin{pmatrix} 1 & \pi \\ 0 & 1 \end{pmatrix}$$

3.)

$$\begin{pmatrix} -4 & 2 \\ 2 & -1 \end{pmatrix}$$

4.)

$$\begin{pmatrix} 3 & 0 & 0 \\ 107 & 1 & 0 \\ \sqrt{2} & 2 & 6 \end{pmatrix}$$

5.)

$$\begin{pmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{pmatrix}$$

6.)

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 1 & 3 & 4 \end{pmatrix}$$

7.) Which of the six matrices from the previous problems have inverses?

Find the inverses of the matrices below.

8.) $\begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix}$

9.) $\begin{pmatrix} 3 & 5 \\ 2 & 3 \end{pmatrix}$

10.) $\begin{pmatrix} 4 & 2 \\ -1 & 3 \end{pmatrix}$

